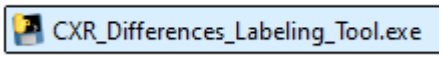
Annotation Software – User Manual

Getting started

Opening the application –

Starting the application by launching the "CXR_Differences_Labeling_Tool.exe" file

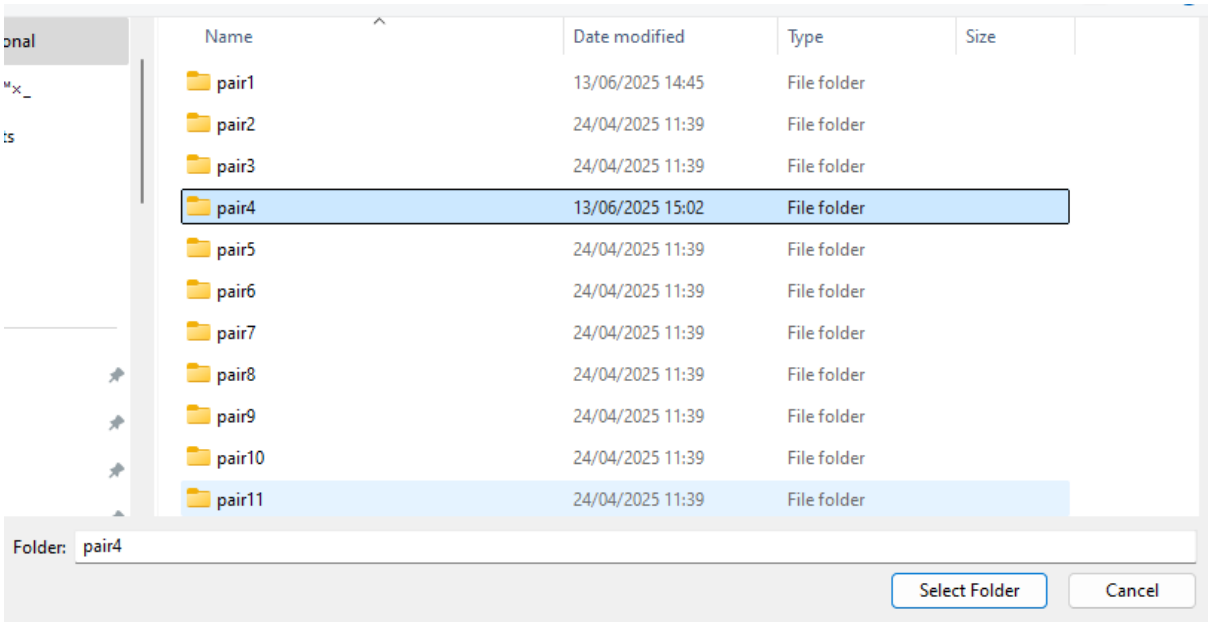


Main window after launch:

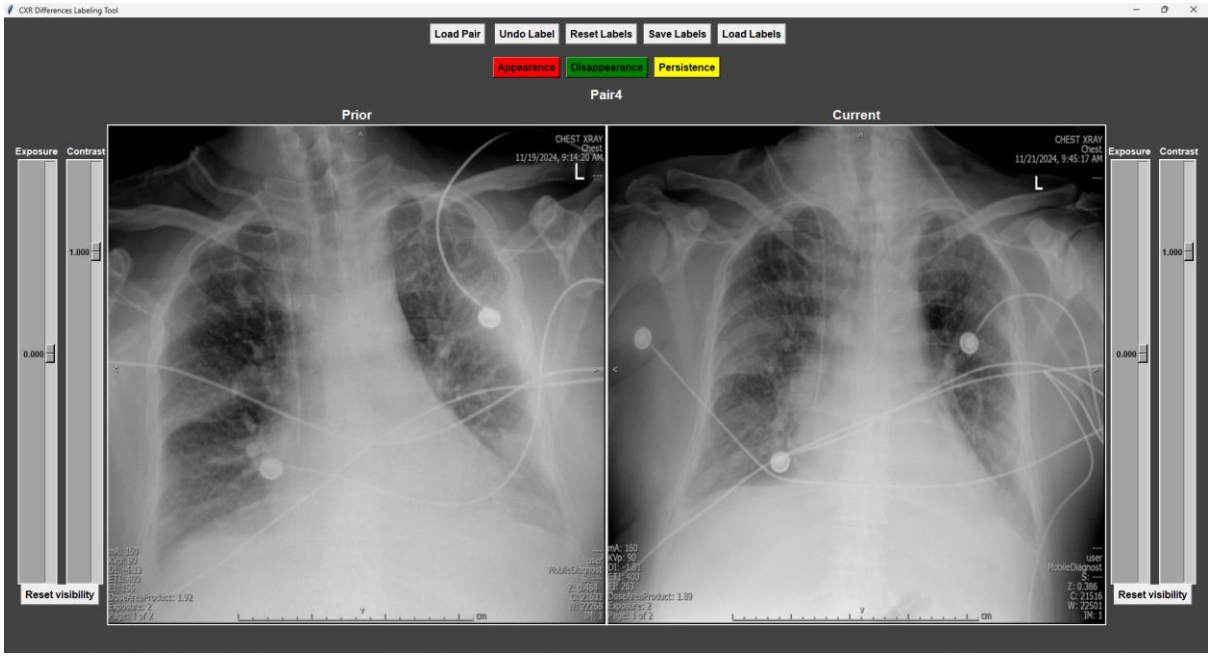
CXR pair loading –

Load a pair by clicking the "Load Pair" button

Navigate to a pair **folder** and select it to load it



View after loading a pair:



The prior image will be on the left, and the current image will be on the right.

Differences Labeling

Choosing a label type –

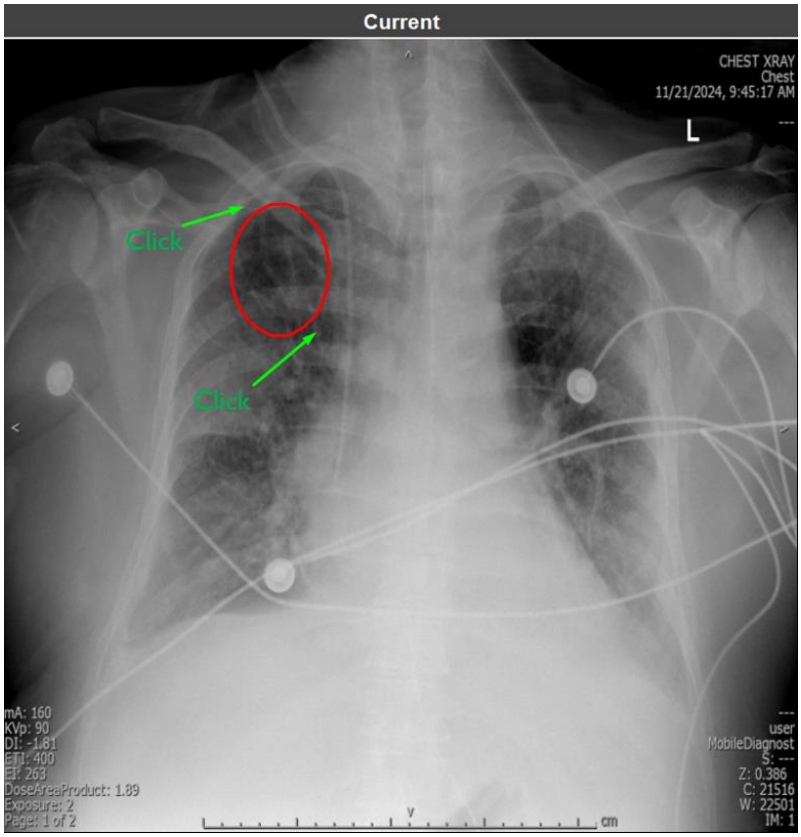
Choose one of the 3 existing label types:

- **Appearance** – For a difference that includes a new entity.
- **Disappearance** – For a difference that includes an entity that has fully healed.
- **Persistence** – For a difference that includes an entity that exists in both the prior and the current.

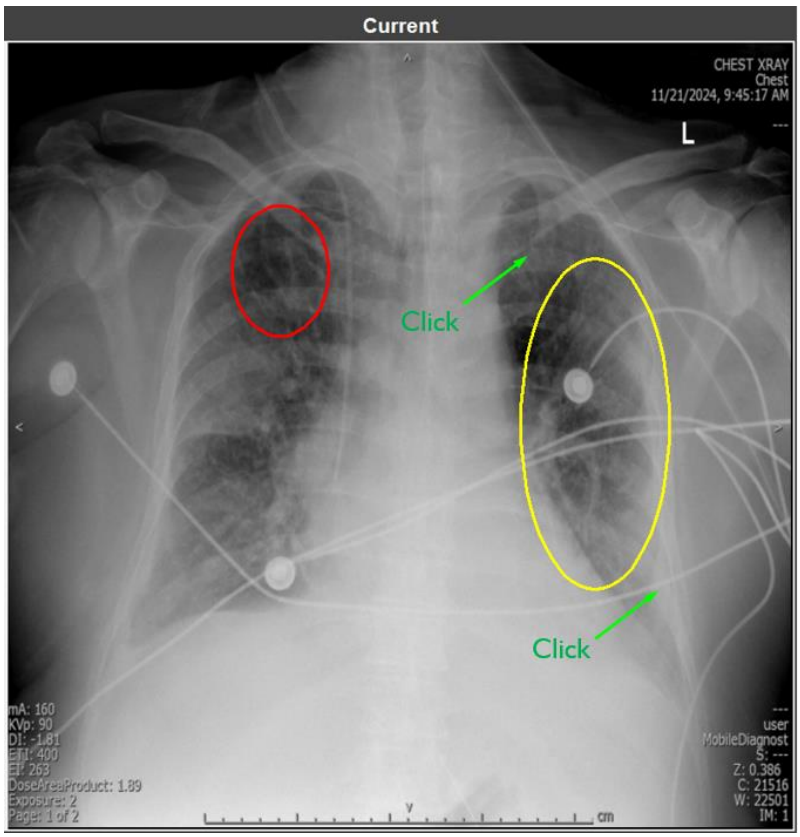
Before placing down a label, click the button that corresponds to the desired label type.

Placing down a label –

To place down a label, click twice on the current:



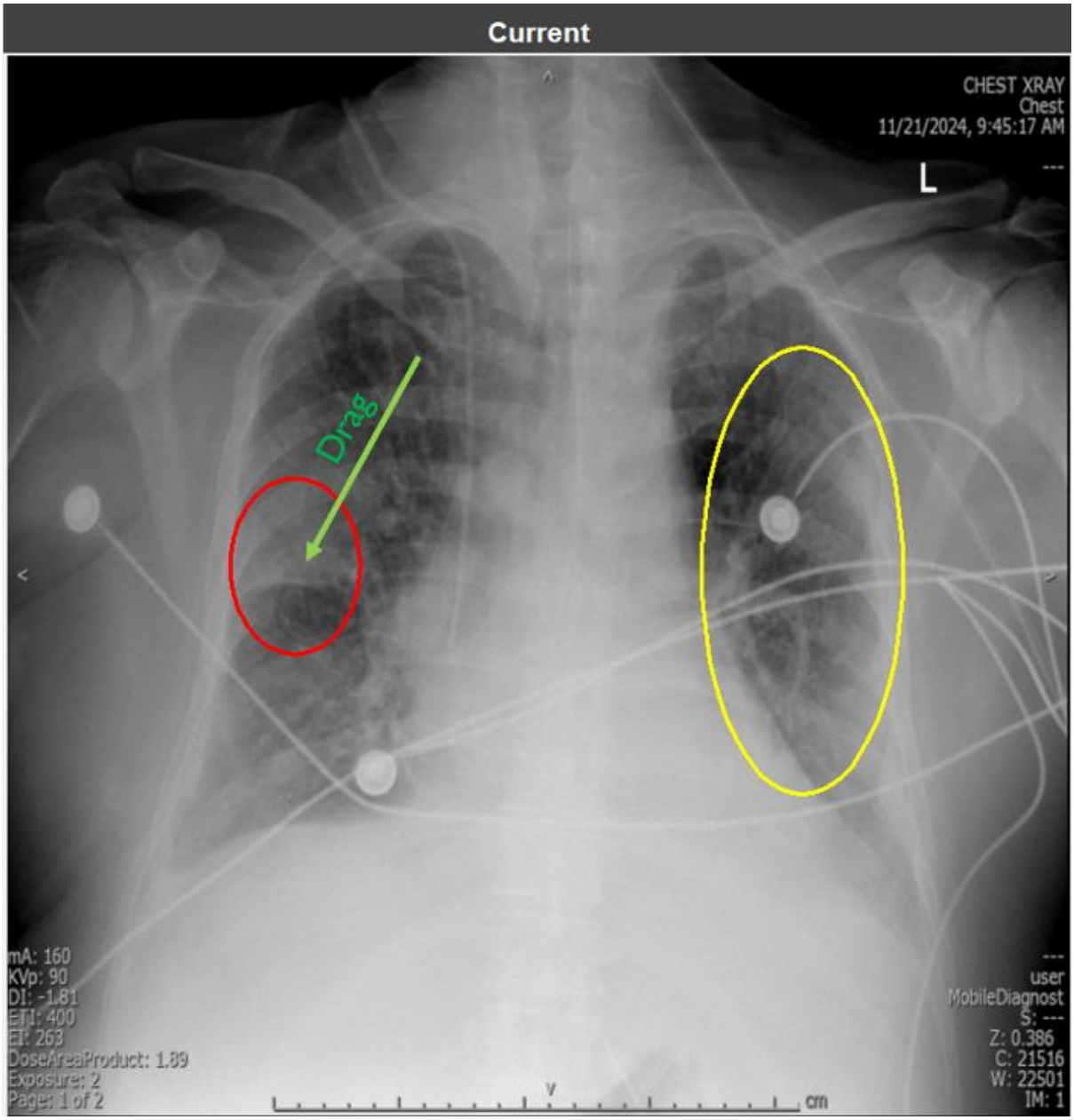
(Label was arbitrarily placed down as an example)



(Label was arbitrarily placed down as an example)

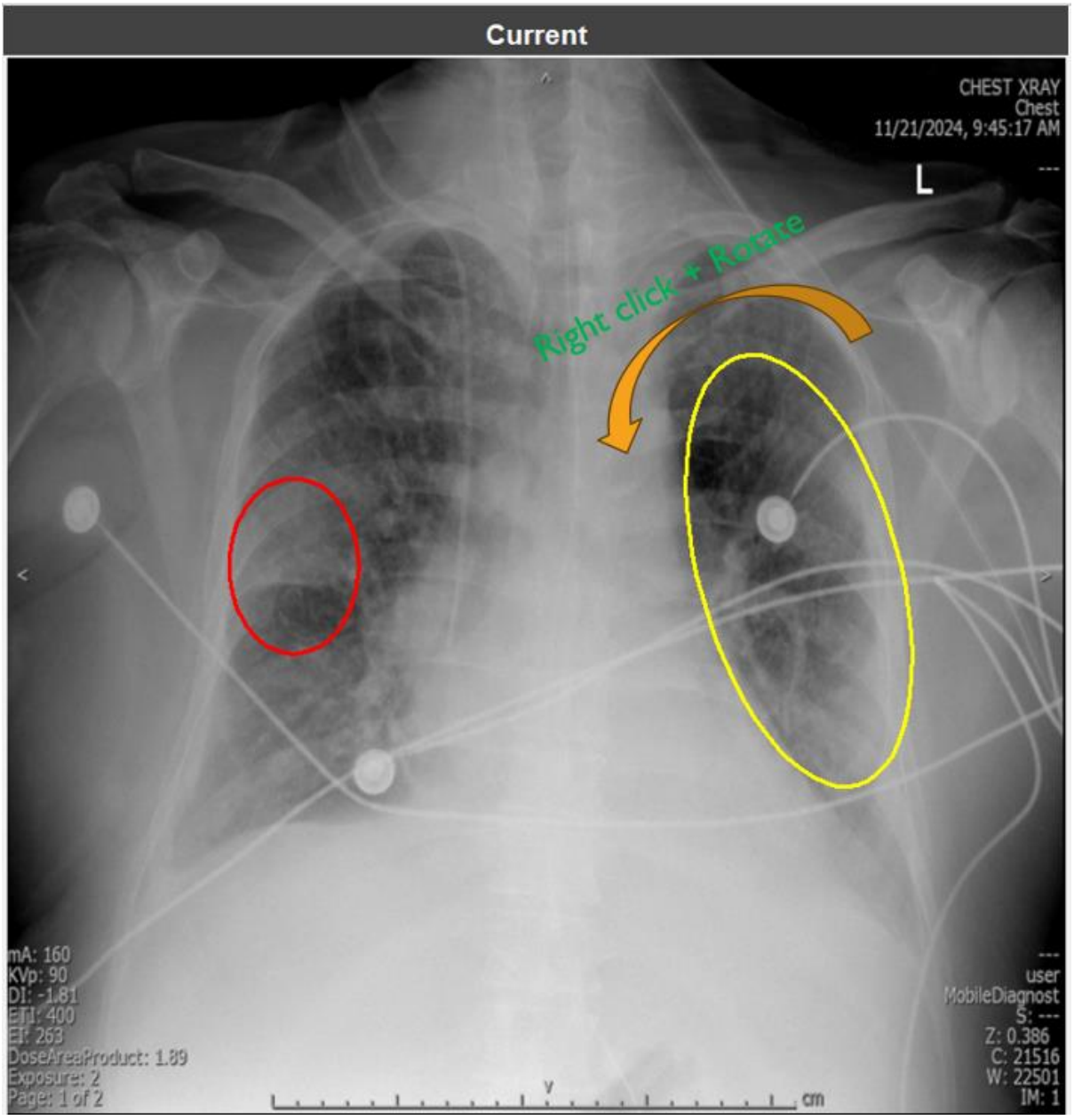
Moving a label –

Move a label after placing it down by dragging it with the left mouse button.



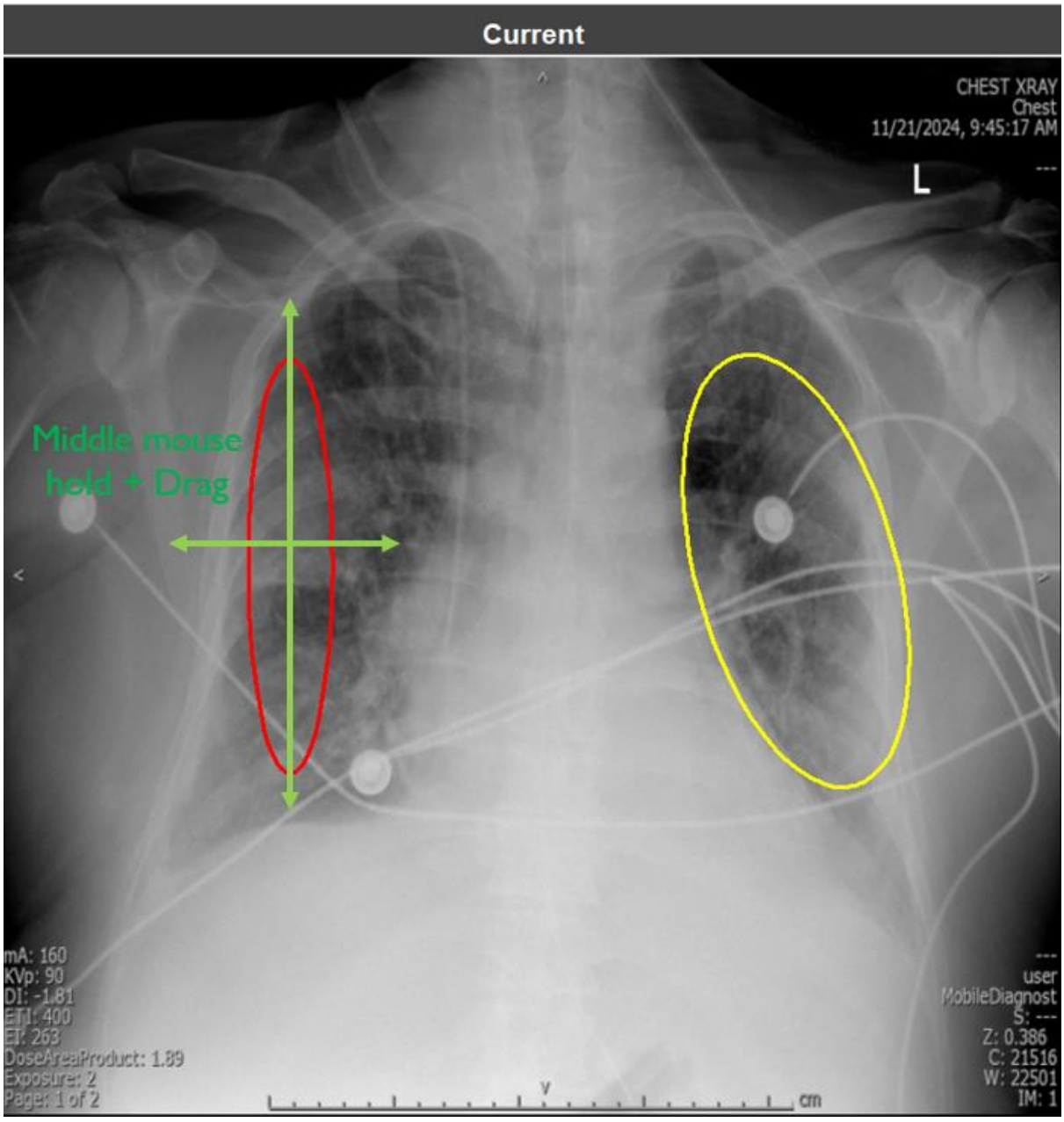
Rotating a label –

Rotate a label by holding the right mouse button inside it and rotating the mouse.



Stretching a label –

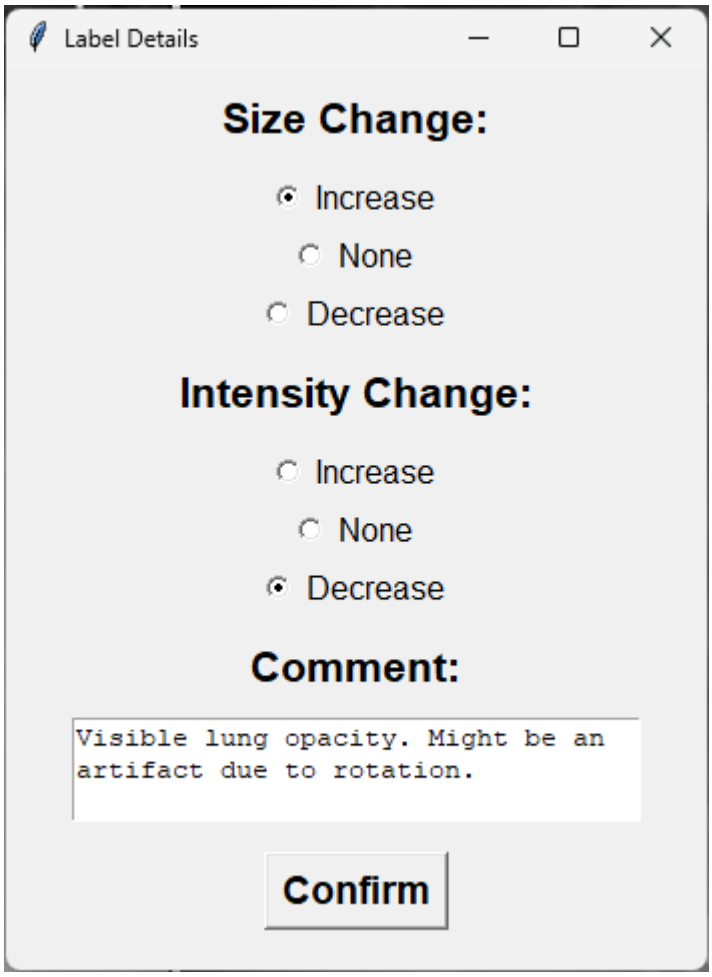
Stretch a label by pressing and holding the middle mouse button (scroll wheel) and dragging the mouse.



Persistence specification –

For every persistence label, the change in size and intensity should be specified. Each can be “Increased”, “Decreased”, or “None”.

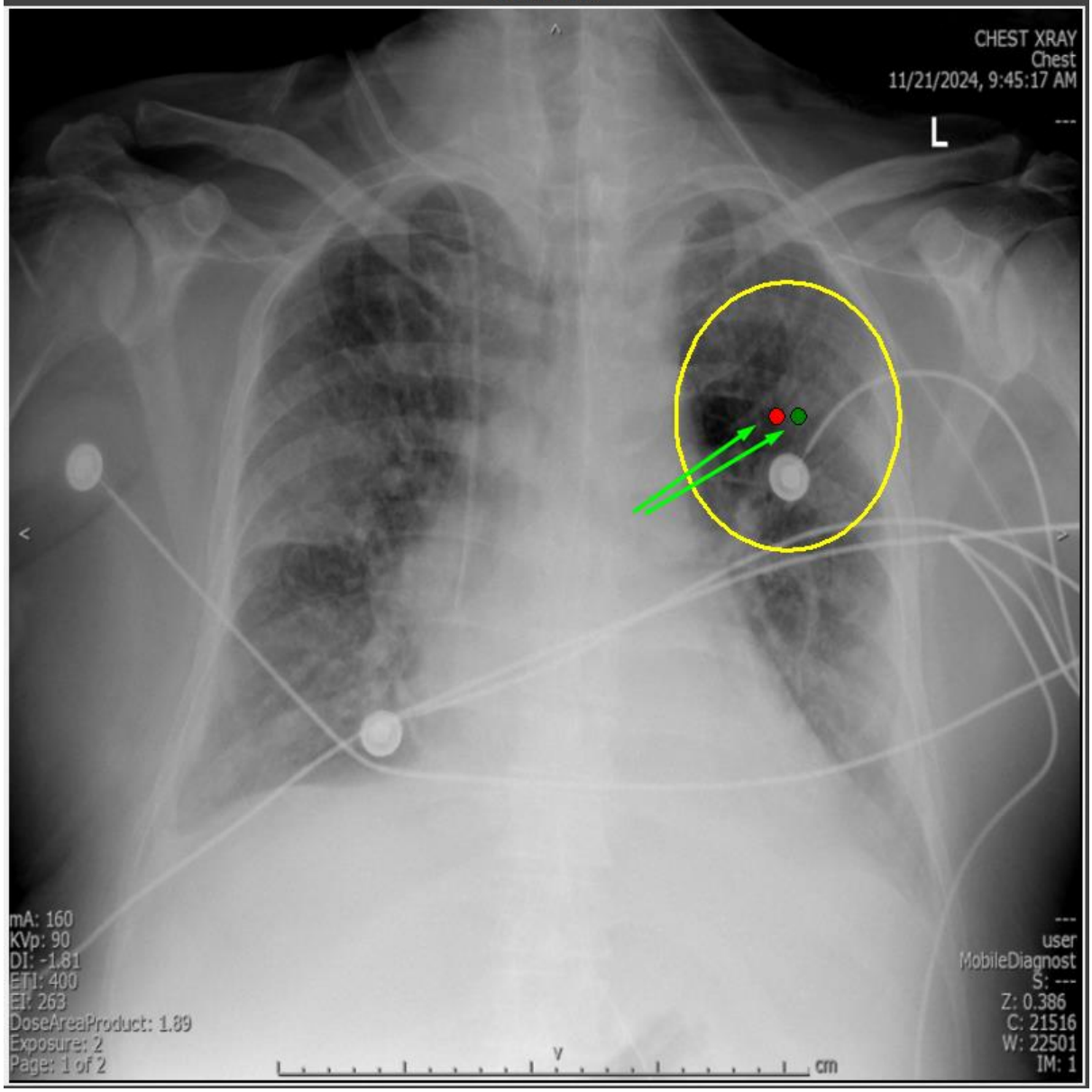
To add a specification for a persistence label (and optionally add a comment), hold the Ctrl button while left-clicking inside a persistence label. A specification window will be opened:



Fill in the details and click “Confirm”.

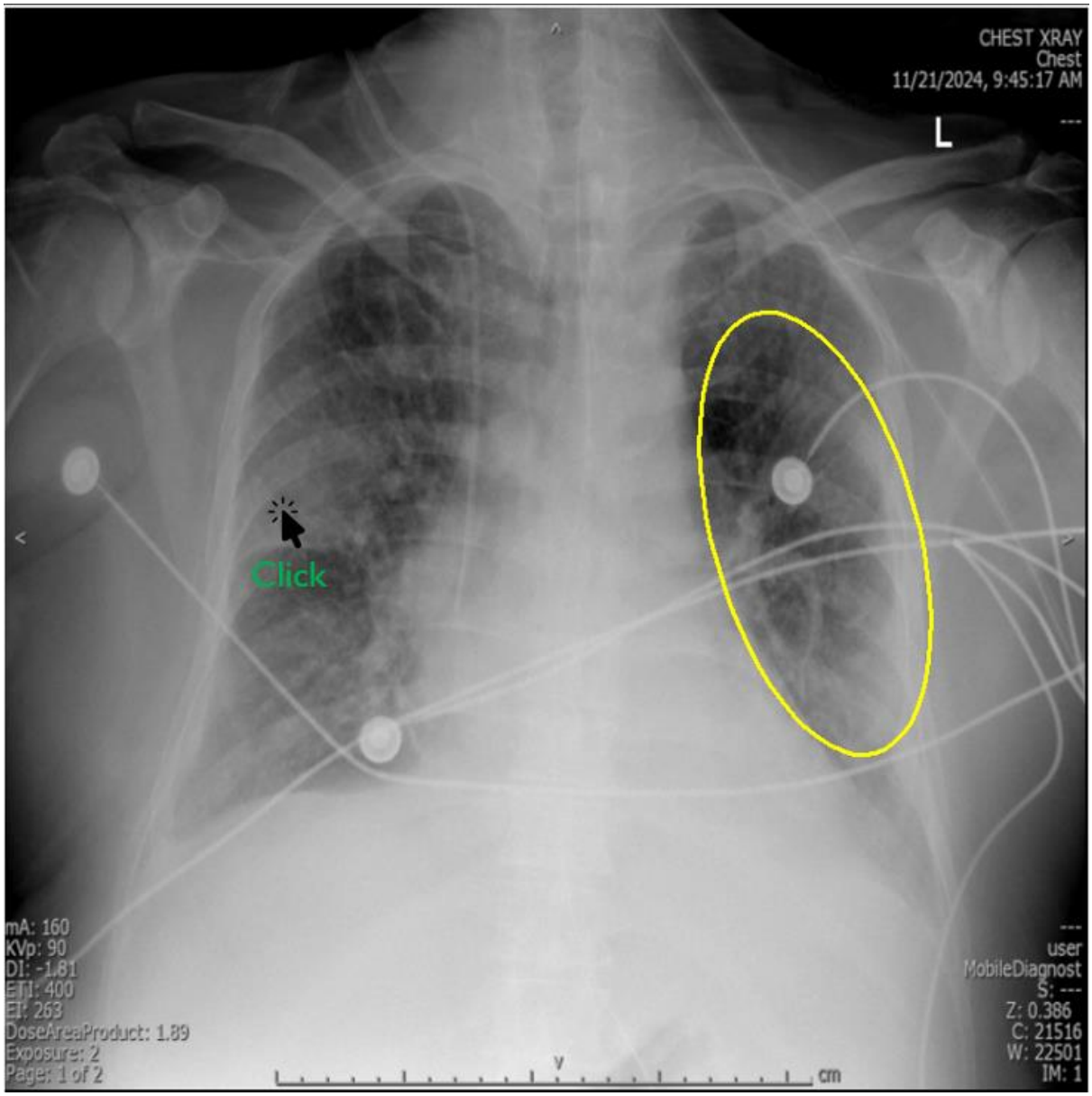
Visual cues that correspond to the specification will be added:

Current


CHEST XRAY
Chest
11/21/2024, 9:45:17 AM
L
---
mA: 160
KVp: 90
DI: -1.81
ETI: 400
EI: 263
DoseAreaProduct: 1.89
Exposure: 2
Page: 1 of 2
cm
---
user
MobileDiagnost
S: ---
Z: 0.386
C: 21516
W: 22501
IM: 1

Label removal

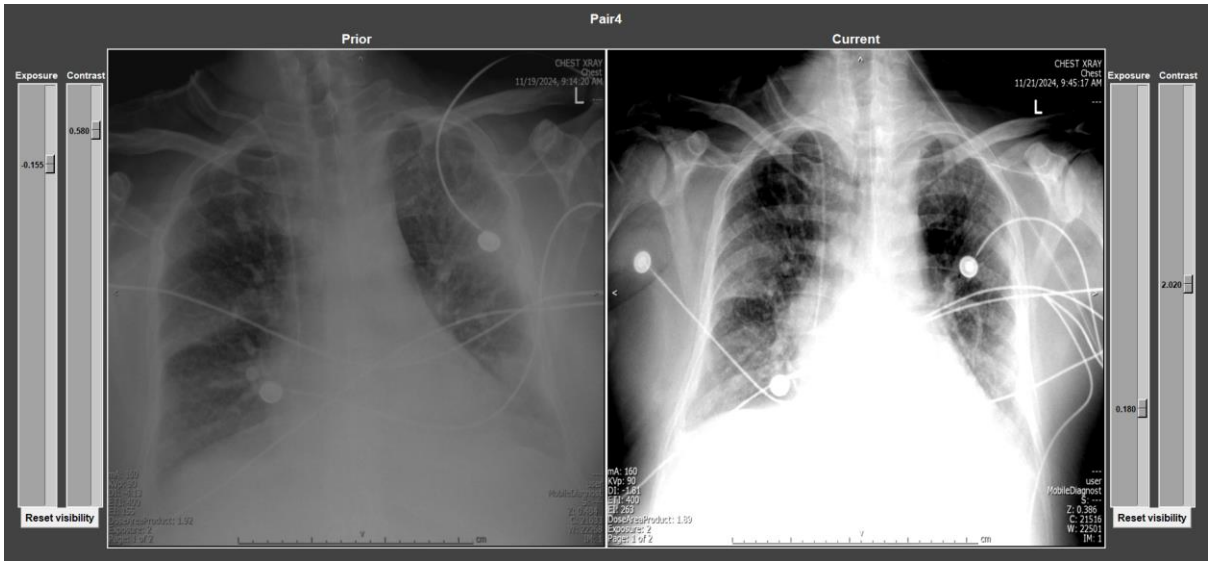
To remove a label, click the “Undo Label” button, and then left-click inside the label you want to remove.



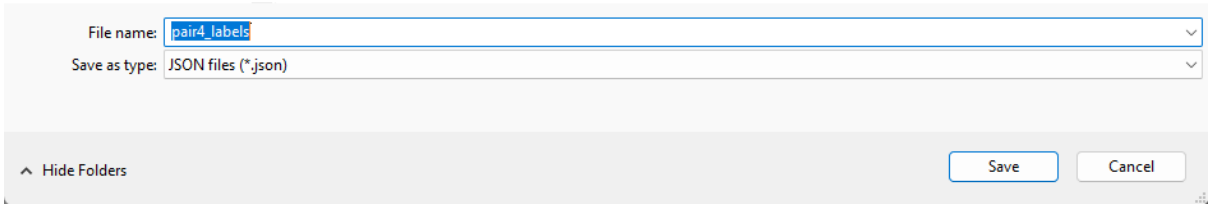
Clicking the “Reset Labels” button removes all currently placed down labels.

Image manipulation

The images' exposure and contrast can be modified using the sliders on each image's sides.



The modifications can be reset by clicking the "Reset visibility" buttons.

Saving labels

To save the currently placed down labels in a file, click the “Save Labels” button, navigate to a desired location, name the file (preferably with a name that corresponds to the pair name), and save. A “.json” suffix will be automatically added.